Day 15: Poisson Distribution
Modeling Count of Events Over Time or Space

100 Days of Data Science

Why Poisson Distribution is Important

Many real-world problems involve counting how many times an event occurs within a fixed interval of time or space.

Examples:

- Number of calls received per minute in a call center
- Number of website visits per hour
- Number of defects per meter of fabric
- Number of accidents at an intersection per month

Poisson distribution provides a mathematical way to model such event counts.

When Do We Use Poisson Distribution?

Poisson distribution is appropriate when:

- Events occur independently
- The average rate (mean) is constant
- We count events in a fixed interval (time, area, length, etc.)

Random Variable

Let X be the number of events occurring in a fixed interval. Then X follows a Poisson distribution with parameter λ (lambda).

Here, λ represents the average number of events in that interval.

Probability Mass Function (PMF)

$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}$$

Where:

- k = number of events (0, 1, 2, ...)
- λ = average rate
- e = Euler's constant (≈ 2.718)

Mean and Variance

For a Poisson distribution:

$$E(X) = \lambda$$

$$Var(X) = \lambda$$

This means the average and variance are equal.

Simple Real-Life Example

A website receives on average 3 visits per minute ($\lambda = 3$).

We can compute probability of exactly 2 visits in the next minute using the formula.

This helps in capacity planning and server scaling.

Business and Operations Example

A factory observes an average of 5 defective items per day.

Using Poisson distribution, managers can estimate:

- Probability of zero defects tomorrow
- Risk of unusually high defect counts

Connection with Binomial Distribution

When number of trials is very large and probability of success is very small, Binomial distribution can be approximated by Poisson distribution.

Example:

- Large number of website visitors
- Very small probability of clicking a rare link

Role in Data Science and Machine Learning

Poisson distribution is used in:

- Modeling event counts (clicks, visits, failures)
- Queueing systems analysis
- Reliability engineering
- Poisson regression (predicting count data)

Example: Predicting number of customer complaints per day.

Interpretation

If λ is higher:

- Events occur more frequently

If λ is lower:

- Events are rare

Common Mistakes

- Using Poisson when events are not independent
- Using it when rate is not constant
- Confusing it with normal distribution for large counts

Final Takeaway

Poisson distribution models how often events occur in fixed intervals.

It is widely used in real-world analytics, operations, and machine learning tasks that involve count data.